

Micro Set 7. Monopolistic Competition & Oligopoly — Model Answer Scheme

Question 1

[2 marks]

Two characteristics from Extract 1.1:

- **Large number of firms each offering differentiated products:** 1810 cafes operating, each with distinctive blends, design, menus, brand identity.
- **Low barriers to entry and exit:** entry cost S\$200000–400000, with new openings (330) and closures (240) keeping the firm count near steady-state.

[1 mark each.]

Examiner Note. Other valid features: substitution available at modest cost; long-run normal profit (~6–8% margin); each firm has some price-setting power through differentiation. The combination of *differentiation* plus *free entry* is the defining mark.

Question 2

[2 marks]

An **oligopoly** is a market structure dominated by a small number of large firms, where each firm's pricing and output decisions significantly affect the others, leading to strategic interdependence. [1 mark.]

Evidence from Extract 1.1: (i) “Three MNOs (Singtel, StarHub, M1) controlled over 96% of the market until 2018” — a small number of large firms; (ii) “Each MNO's pricing decision is highly interdependent. . . if Singtel cuts mobile data prices, StarHub and M1 must respond within weeks” — strategic interdependence. [1 mark.]

Question 3

[4 marks]

Product differentiation and downward-sloping demand.

In monopolistic competition, each firm produces a product distinguishable from its competitors — by recipe, atmosphere, location, brand, or social-media presence (Extract 1.1: “We are competing not on coffee quality alone. . . but on atmosphere, location, social media presence, and brand identity”). This differentiation gives each cafe some degree of consumer loyalty: customers willing to pay a small premium for their preferred cafe rather than substituting freely. [2 marks for differentiation → consumer loyalty.]

Because of this loyalty, raising price by a small amount does *not* cause customers to abandon the cafe entirely (as would occur in perfect competition, where products are homogeneous). Instead, customers continue patronising the cafe, with only a moderate fraction switching to alternatives. The firm faces a downward-sloping demand curve: it can raise price and still retain some demand, but loses progressively more demand as price rises. [2 marks for the price-quantity mechanism.]

Conversely, in perfect competition where products are homogeneous, any single firm raising price above the market level would lose *all* its customers immediately — a horizontal (perfectly elastic)

demand curve. Differentiation breaks this symmetry. *[1 mark for contrast (bonus point typically credited within the 4 marks).]*

Examiner Note. The strongest answers link *differentiation* to *partial loyalty* to *less-than-perfectly-elastic demand*. Pure assertion (“differentiation = downward-sloping demand”) without the loyalty step earns 2 marks.

Question 4

[4 marks]

Why oligopolists compete on non-price factors.

Extract 1.1 describes the pricing interdependence: “If Singtel cuts prices, StarHub and M1 must respond within weeks — or risk losing subscribers en masse.” This means that price competition is largely matched by rivals, eroding any individual MNO’s advantage. *[1 mark for matched-response logic.]*

The result is that price cuts trigger a price war without permanently gaining market share, and price rises are not followed by rivals — creating an asymmetric loss potential. The optimal strategy for an oligopolist is to avoid price competition and compete instead on dimensions where rival response is delayed or less effective. *[2 marks for asymmetric-response argument.]*

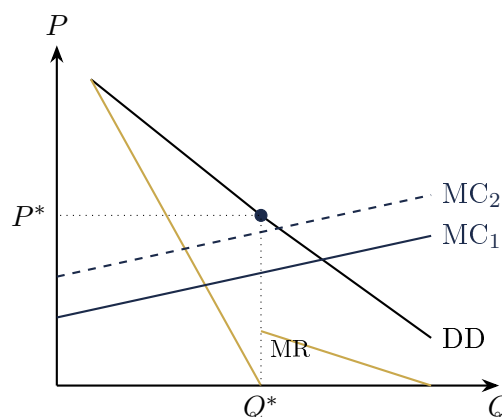
Non-price competition — network coverage, 5G rollout speed, bundled services, brand prestige (Extract 1.1) — has three advantages: (i) rival response is slower (network investment takes years), (ii) differentiation creates customer lock-in, and (iii) the dimensions are harder for rivals to directly match. This allows MNOs to maintain margins while still competing intensely. *[1 mark for non-price advantages.]*

Examiner Note. The kinked-demand-curve intuition underlies this answer: price cuts are matched (steep response), price rises are not (shallow response). Connecting this to “non-price” competition is the key analytical move.

Question 5

[6 marks]

Kinked demand curve and price stability.



In the diagram, the demand curve is kinked at the prevailing price P^* . Above P^* , the demand curve is elastic: if the MNO raises price, rivals do *not* follow (fearing market-share loss), so the MNO loses substantial market share. Below P^* , the demand curve is inelastic: if the MNO cuts

price, rivals match immediately, so the MNO gains only proportionate market share. *[2 marks for diagram + kink interpretation.]*

The marginal revenue (MR) curve corresponds to the kinked DD: it has a discontinuous vertical gap at Q^* , reflecting the change in slope of demand. *[1 mark for MR discontinuity.]*

The crucial insight: the MNO's profit-maximising output occurs where $MC = MR$. As long as the firm's MC curve passes through the MR discontinuity (between MC_1 and MC_2 in the diagram), the profit-maximising output and price do *not* change. The firm can absorb significant changes in marginal cost (e.g. from changing network operating costs or wholesale rates) without altering its price. *[2 marks for MC-shift price stability.]*

This explains why oligopolistic prices tend to be “sticky” — they remain stable even as underlying cost conditions vary, until cost shifts outside the MR discontinuity range. *[1 mark for price-stickiness conclusion.]*

Examiner Note. The kinked-demand model's core insight is the MR discontinuity. Without explicitly drawing and explaining it, the analysis is incomplete.

Question 6

[6 marks]

MVNO entry and pricing disruption.

Reason 1: Breaking the established equilibrium. Before MVNO entry, the three MNOs held over 96% combined market share and maintained tacit pricing coordination. MVNOs entered with significantly lower retail prices (mobile data fell from S\$4.50/GB in 2019 to S\$0.80/GB in 2024 — 82% decline). This entry shifted the kinked demand curve faced by each MNO: above the new lower price, demand is now even more elastic because MVNOs offer a credible substitute. The price equilibrium therefore moved down sharply. *[2 marks: kink-shift mechanism.]*

Reason 2: Lower-cost business model. MVNOs rent capacity from MNOs at wholesale rates rather than owning infrastructure, giving them a leaner cost structure. This makes them credible long-run competitors at low prices — they can sustain price cuts that incumbent MNOs find harder to match given their fixed-cost-heavy infrastructure. The result is genuine downward pressure on prices rather than reversible price wars. *[2 marks: cost-structure asymmetry.]*

Reason 3: Segmentation of demand. MVNOs targeted price-sensitive segments (younger users, occasional users, secondary SIMs). This removed the most elastic share of demand from the MNOs, leaving MNOs to compete primarily for the less-elastic core segments. MNOs responded with no-frills sub-brands (Singtel's GoMo, StarHub's Giga!, M1's MyRepublic) to recapture this segment, intensifying competition. By 2024, MVNO + MNO sub-brand combined share reached 25–30% of the market, fundamentally altering the structure from a three-firm oligopoly toward a more fragmented landscape. *[2 marks: segmentation + structural change.]*

Examiner Note. Strong answers identify that MVNO entry changed not just price but *market structure*. Listing only “prices fell because of competition” captures the symptom but not the structural cause.

End of Set 7 — Model Answers